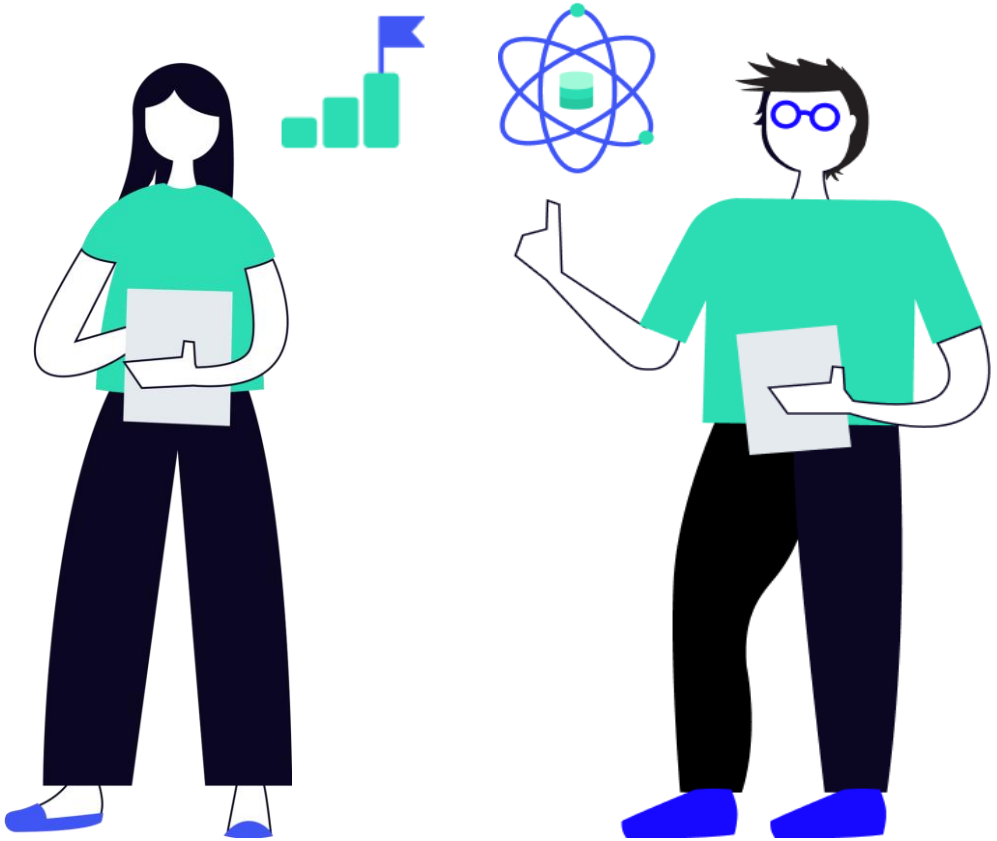
Dilemma in ML development

Building everything **manually** from scratch vs. using a **tool to support** the development phase (from collecting data to deploying on the edge).



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Main pain points in ML workflows

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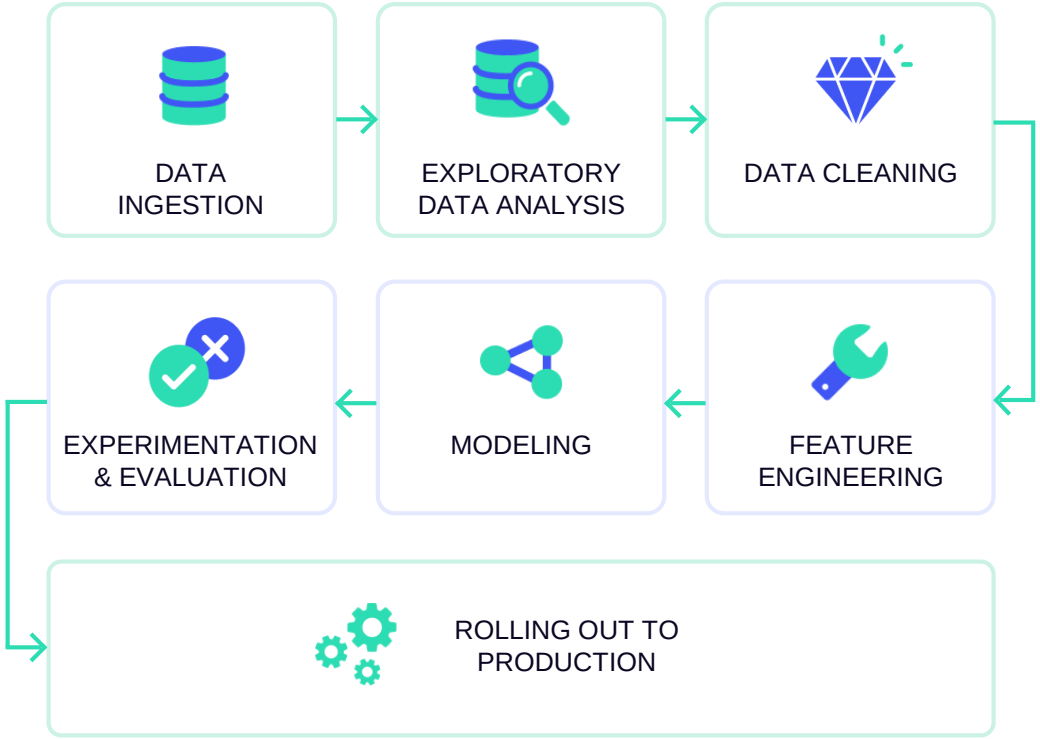
Standard ML workflow



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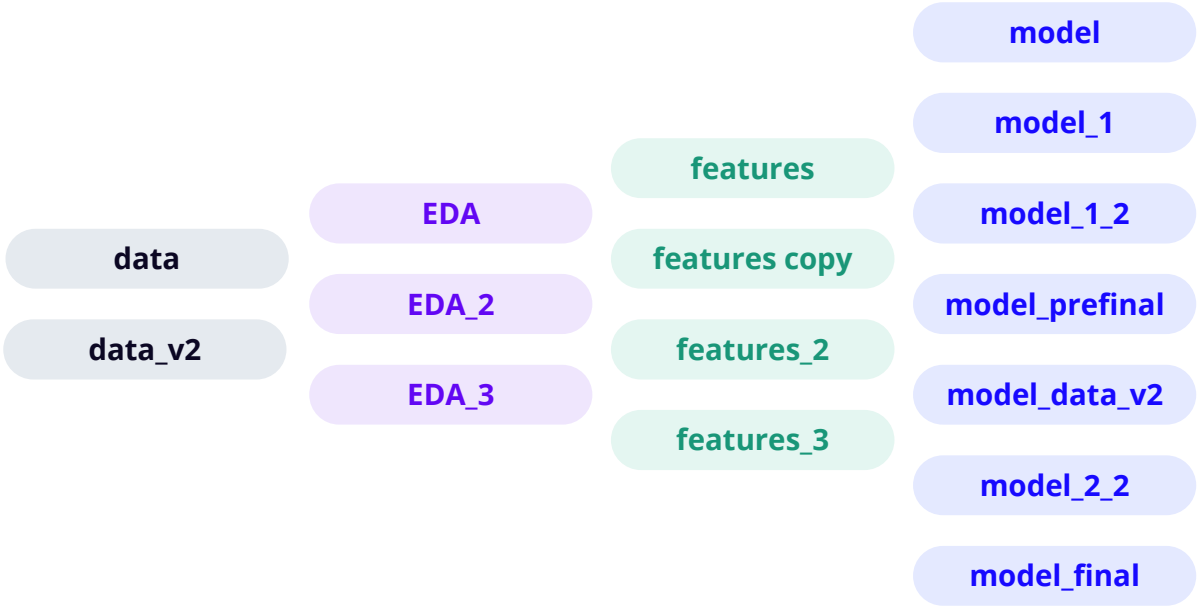
ML pipeline in practice



*EDA = Exploratory data analysis

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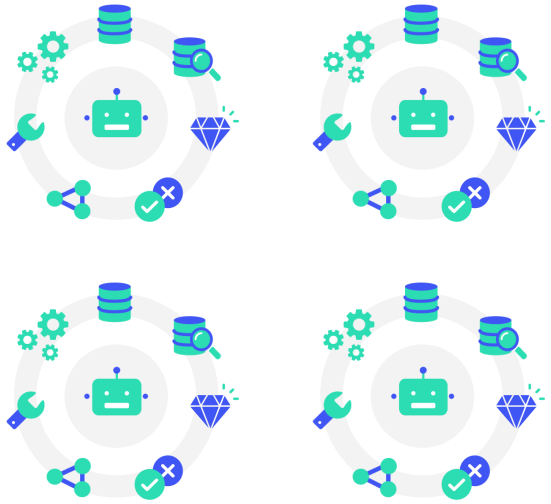
Main pain points in ML workflows



1. Reproducibility

- Teamwork
- Usually ad-hoc processes
- Productivity bottleneck
- Challenges
  - Changes in data
  - Hyperparams inconsistency
  - Randomness
  - Manual and ad-hoc execution of experiments

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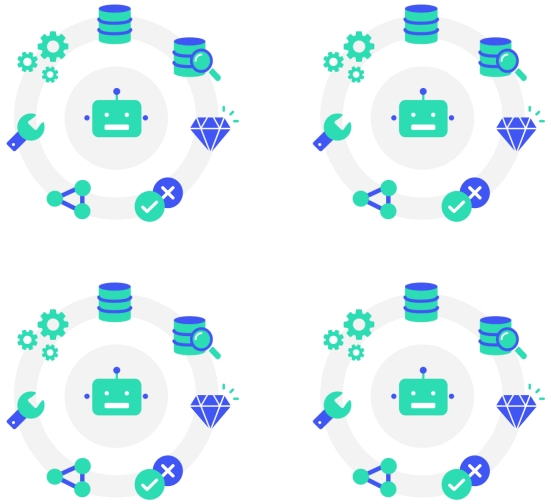
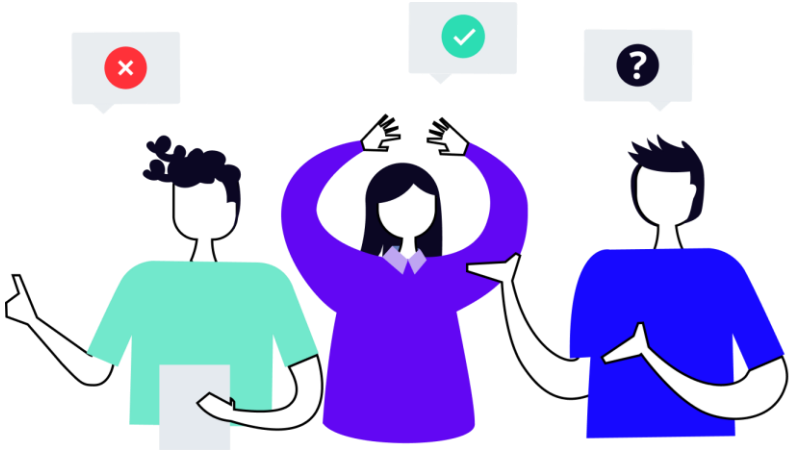
Main pain points in ML workflows



1. Reproducibility



"Changes are uploaded, please run all the notebook again."

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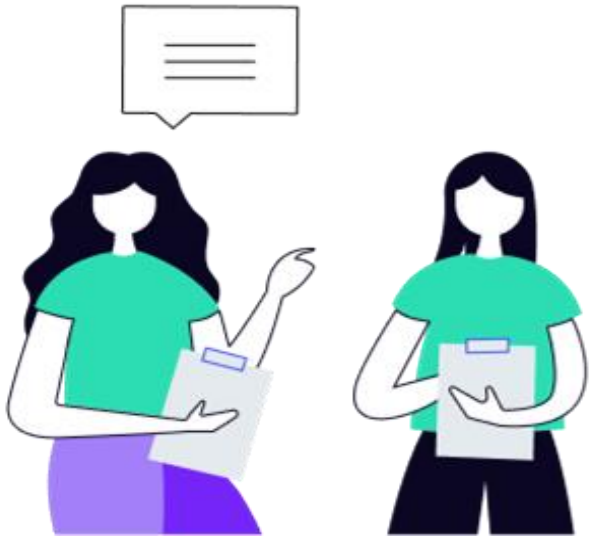
Main pain points in ML workflows



2. Data sharing

- Complex READMEs on how to gather data from remote storage
- Security and data privacy risks
- Manual versioning of dataset changes

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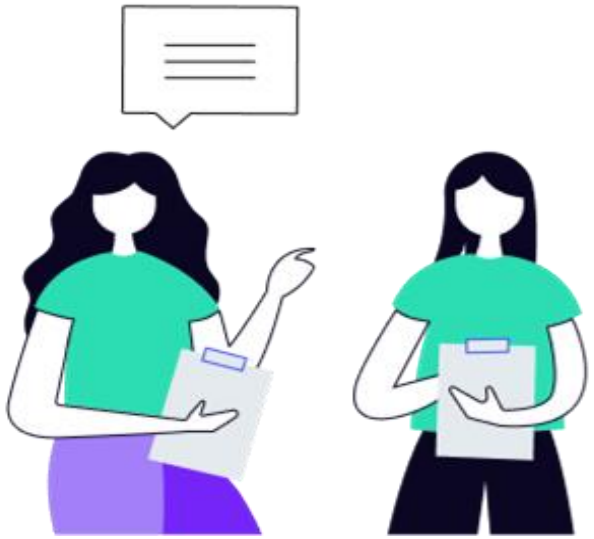
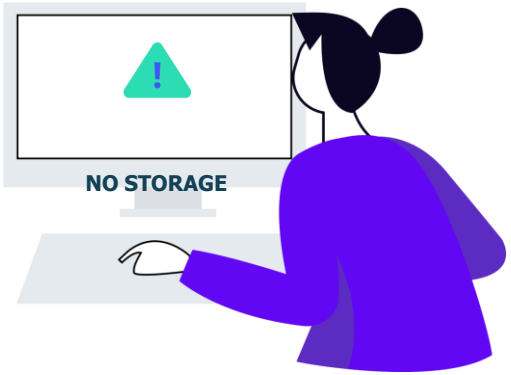
Main pain points in ML workflows



2. Data sharing



"I wish I could automate this process…"

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Main pain points in ML workflows

3. Experiments execution & tracking

- Experiments setup traceability challenges
- Inefficient results comparison & evaluation
- Manual process:
  - Spreadsheet
  - Github (metadata files)
  - Tracking tools (big learning curve)

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Ideal development experience

Structured pipeline composed by interdependent steps

Easily **adding files or directories** to a remote repository

Sharing experiments, models, and results in a simple way

Stop worrying about source code and data association

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Useful open source tool:

DATA VERSION CONTROL

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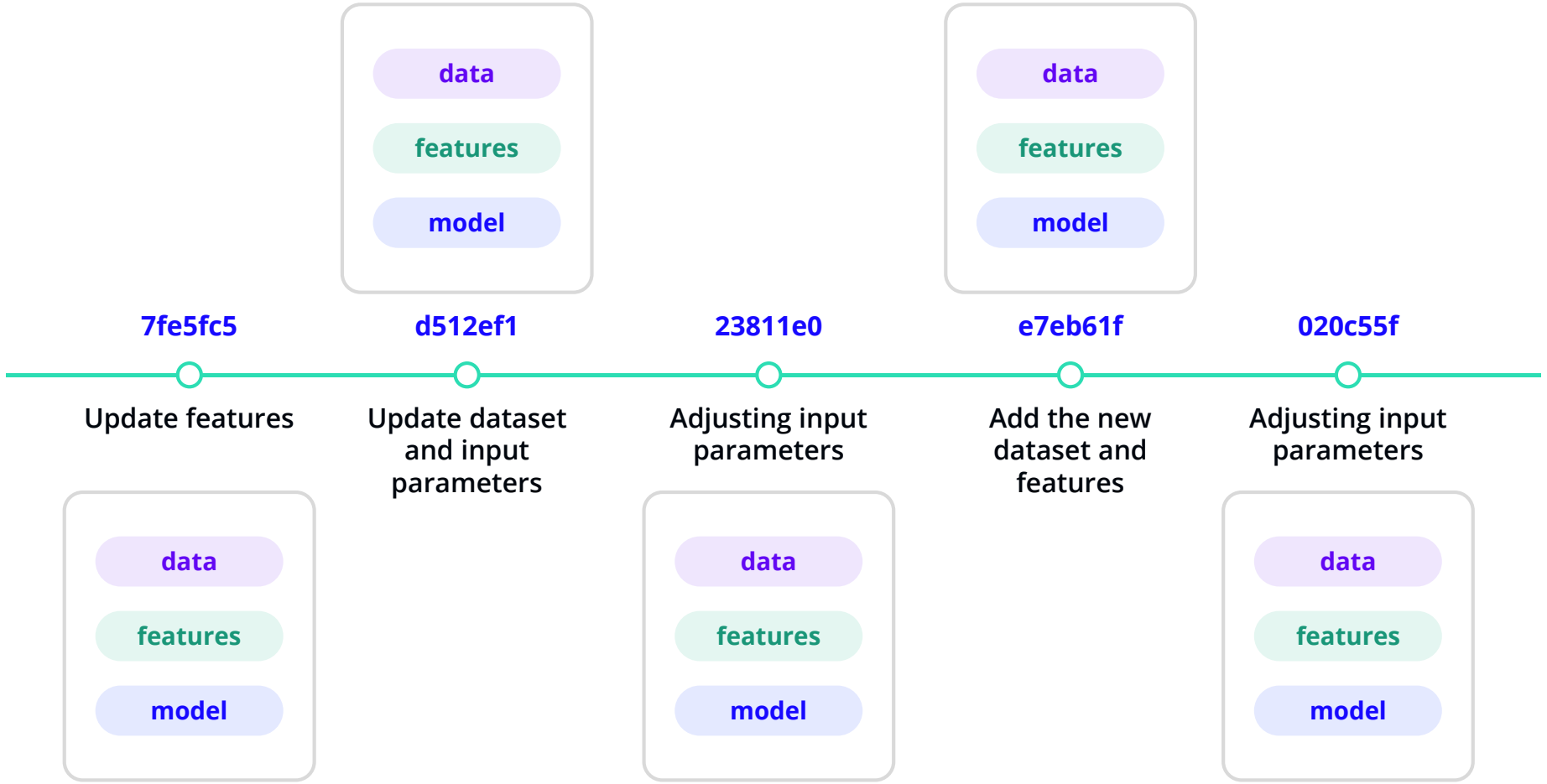
DVC high-level overview



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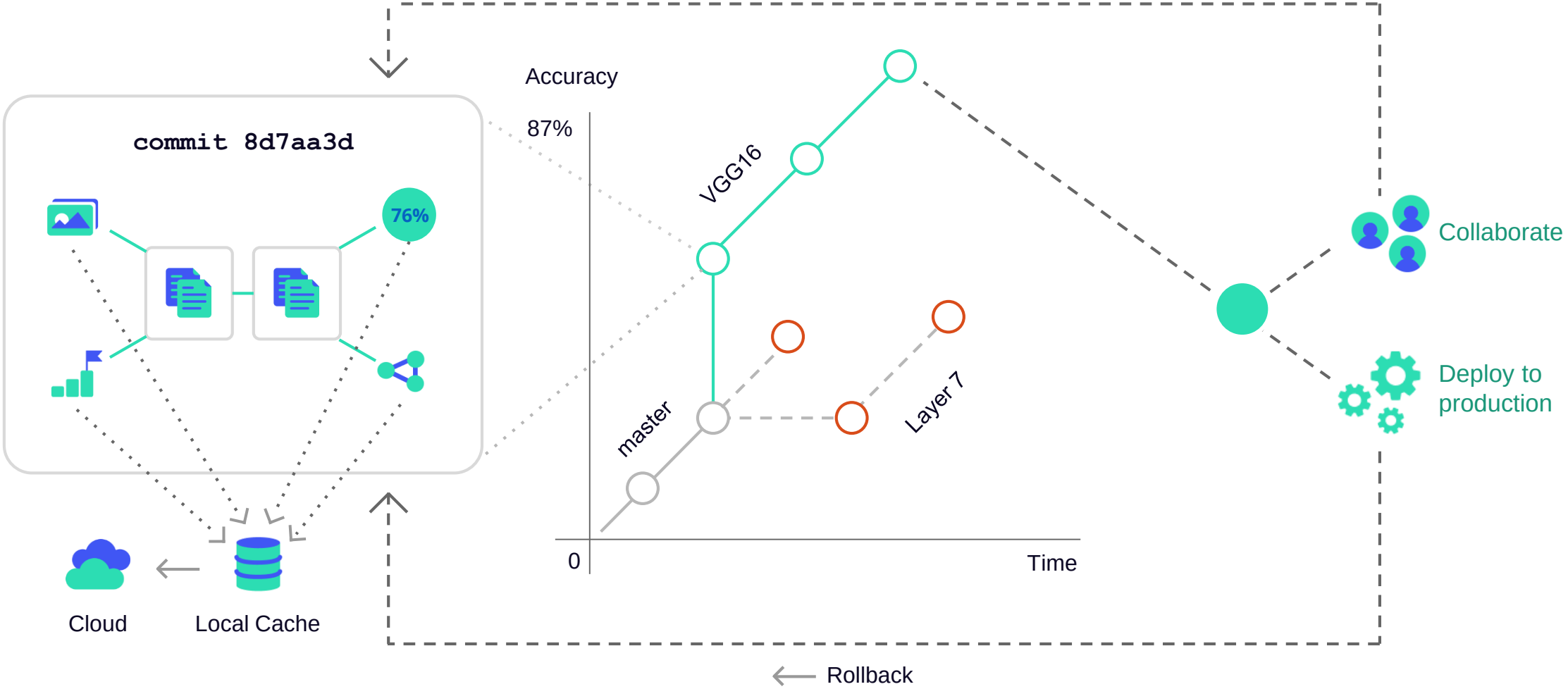
DVC high-level overview



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Main features

- Git-compatible
- Storage agnostic
- Reproducible
- Low friction branching
- ML pipeline framework
- Language & framework agnostic
- Track failures
- Experiments & metrics tracking

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Pipelines

- **Pipelines** composed by interdependent **steps**
  - Dependencies
  - Code to execute
  - Outputs

- Additional pipeline **visualization** command `dvc dag`

```
$ dvc dag
          +---------+
          | prepare |
          +---------+
               *
               *
               *
        +-----------+
        | featurize |
        +-----------+
         **        **
       **            *
      *               **
+-------+               *
| train |             **
+-------+            *
         **        **
           **    **
             *  *
        +----------+
        | evaluate |
        +----------+
```

```yaml
stages:
  build:
    cmd: python train.py
    deps:
      - features.csv
    outs:
      - model.pt
    metrics:
      - accuracy.txt:
          cache: false
    plots:
      - auc.json:
          cache: false
```


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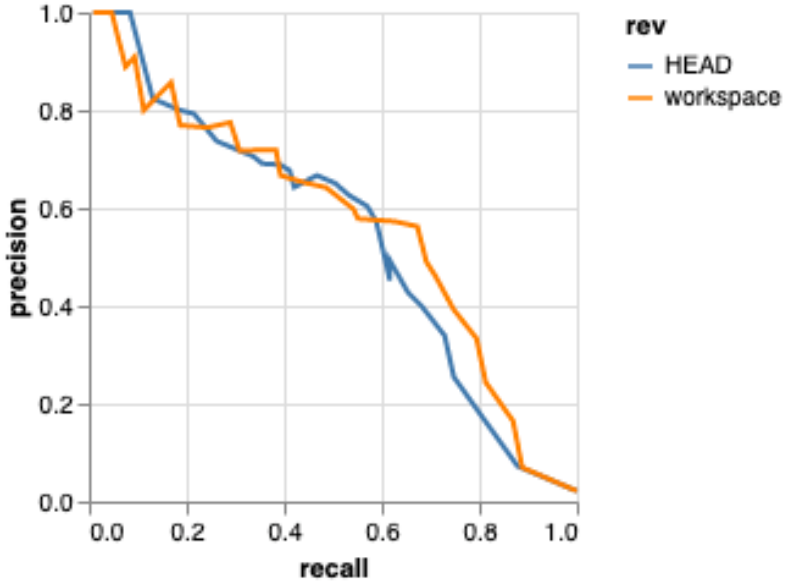
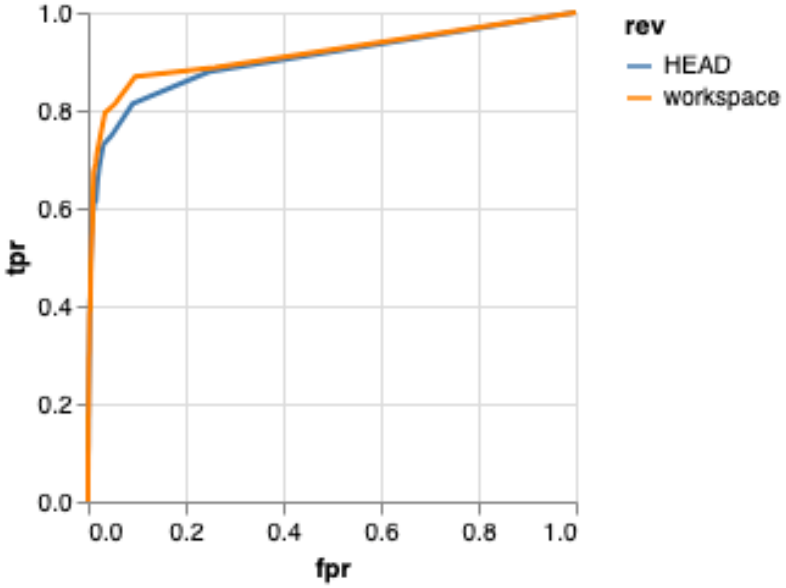
Metrics differences

Smooth comparison process:
numeric and **graphic** visualization

```
$ dvc metrics diff
Path          Metric    HEAD     workspace  Change
scores.json   avg_prec  0.52048  0.55259    0.03211
scores.json   roc_auc   0.9032   0.91536    0.01216
```





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Continuous integration

- Automatically check data version
- Benchmark new model against previously deployed models
- Metrics diff & interactive plots in Pull Requests
- Re-train & refine in the cloud

SOURCE: WWW.DVC.COM

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Experiments batch execution

Experiment	Created	train	test	model.n_estimators	model.max_depth	model.min_samples_split	model.min_samples_leaf	model.max_leaf_nodes	model.random_state
workspace	-	96.257	70.404	100	20	2	1	-	42
dvc	02:59 PM	96.257	71.3	100	-	2	1	-	42
├─ 20d798f [max_depth_20]	03:49 PM	96.257	70.404	100	20	2	1	-	42
├─ 6d9edfa [max_depth_5]	03:49 PM	76.946	74.439	100	5	2	1	-	42
├─ ac459ee [max_depth_1]	03:49 PM	68.263	71.749	100	1	2	1	-	42
└─ a7acf28 [max_depth_2]	03:48 PM	71.557	76.233	100	2	2	1	-	42

"I can't believe the number of hours saved by queuing and executing experiments in parallel."

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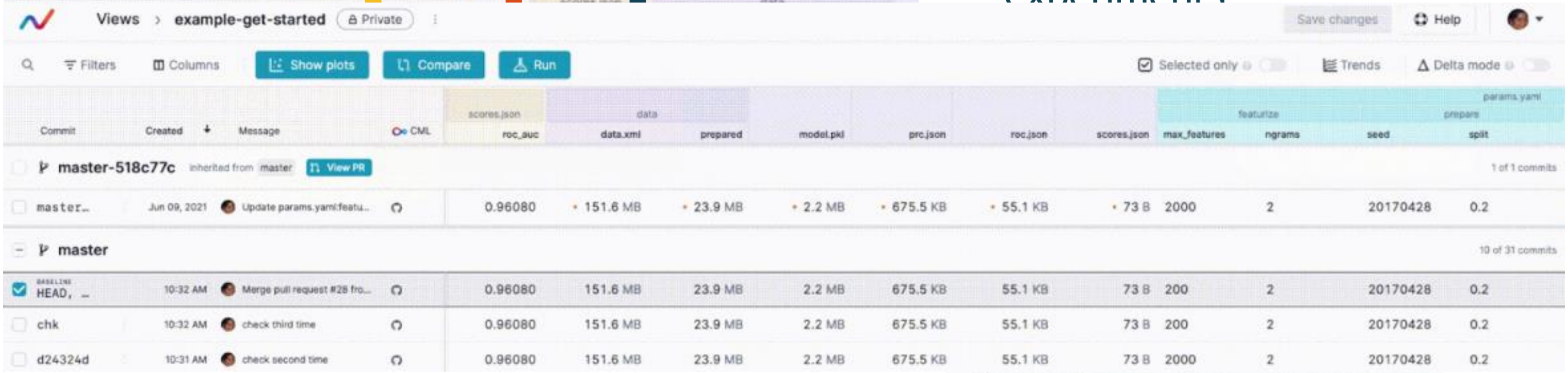
UI does not have to be built from scratch

- Show plots for selected experiments



- Generate trend charts

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